

MADHURA PRATAP HANDE DESHMUKH

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SUMMARY

Embedded Software Engineer delivering high-impact solutions across medical, robotics and embedded systems. Proven success designing real-time firmware, signal processing algorithms that enhance performance, reduce downtime, shorten launch cycles. Driven to create embedded innovations that lower cost, boost reliability and fuel business growth.

EXPERIENCE

Noccarc Robotics Pvt Ltd

Bhosari, Pune

Embedded Software Engineer

June 2024 – present

- Developed firmware modules for signal acquisition, preprocessing, vital parameter computation in Patient Monitor NX5—a critical care platform projected to generate ₹8–10 crore annually, supporting Noccarc's expansion in ICU, hospital market.
- Delivered embedded algorithms for physiological monitoring (NIBP, ECG, Respiration, Temperature, Capnography) with 98% accuracy, reducing processing latency by 20% while conforming to IEC80601, ISO80601, and IS13450 standards.
- Authored comprehensive Firmware Requirements Specifications (FRS) and systematically structured detailed system architecture, strengthening documentation clarity, design traceability, and regulatory compliance readiness.
- Ensured 100% MISRA-C/C++ compliance across 20K lines of code, passing internal audits without violations.
- Integrated ICU clinician feedback to refine algorithms - improving measurement consistency >90% across ICU trials.
- Diagnosed and fixed battery management bugs in Ventilator V730i firmware; cutting ICU device downtime by 40%.

Noccarc Robotics Pvt Ltd

Bhosari, Pune

Embedded Software Intern

October 2023 – May 2024

- Designed and integrated Volumetric Capnography Analysis for Ventilator V730i, enabling real-time monitoring of patient metabolism, respiratory health, and CO₂ exchange—vital for ICU decision-making and improved outcomes.
- Enhanced real-time waveform filtering algorithms for ECG and Capnography, improving signal-to-noise ratio by 35%.
- Automated test procedures for two critical PCB validation fixtures, reducing manual testing time from 25 minutes to 4 minutes per unit and improving Ventilator V730i production line efficiency by 70%.

TECHNICAL SKILLS

Programming: C / C++, Embedded C, Python

Microcontrollers: STM32 (HAL, LL), MSP430, ATmega (2560, 328P)

Operating Systems & RTOS: Linux (Ubuntu), Windows, FreeRTOS, CMSIS-RTOS, bare-metal development

Communication Protocols: UART, SPI, I2C, CAN, RS485, One-wire, USB, Bluetooth, Wi-Fi, Zigbee

Interfaces: ADC, DAC, DMA, PWM, Timers, GPIO, External Interrupts

Development Tools: Keil uVision, Code Composer Studio, Eclipse IDE, VS Code, Git, Eagle, Yocto

Debug & Test Equipment: JTAG, ST-Link, Oscilloscope, Logic Analyzer, Function Generator, Multimeter, DC Load

Design & Integration: PCB Design, Motor Drivers (DC, BLDC, Stepper), Pneumatic Actuators, PID Control, Control Systems

Languages: English, Hindi (working proficiency), Marathi (native)

ACADEMIC PROJECTS

Design and Development of Balancing Robot

October 2022 - May 2023

- Built a self-balancing robot on a spherical wheel with 6-DOF IMU, PID control; stabilized <2s from external disturbances.
- Operated with $\pm 2^\circ$ tilt accuracy; performance validated across diverse surfaces and disturbance scenarios.

Robotic Hand Prosthesis

August 2021 - September 2022

- Developed an EMG-based prosthetic hand with 3 DOF, capable of recognizing 5 distinct muscle activation patterns.
- Enabled lifelike finger articulation with <150ms response time, aiding intuitive control for users in simulation trials.

EDUCATION

Government College of Engineering and Research, Savitribai Phule Pune University

Pune

Bachelor of Engineering in Instrumentation and Control

August 2019 - June 2023

- CGPA: 8.74/10

AWARDS & HONORS

- University Rank 10 — Instrumentation & Control Branch, SPPU (2023)
- 2nd Place — COEP Tech I2I Innovation Challenge (2022) for Robotic Hand Prosthesis
- All India Rank 6 — ABU Robocon 2021, All India Rank 14 — ABU Robocon 2020